Verify-or-Abstain Evidence Pipelines: When Cryptographic Custody Is Necessary but Not Sufficient

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Abstract

Scientific evidence pipelines increasingly rely on content-addressed custody and hash-valid artifacts, yet syntactically valid objects can remain semantically inconsistent. We present an evidence-pipeline reliability study centered on explicit terminal states, row-level accounting invariants, and closed-vocabulary identity guards. Preregistered audits show that (i) a historical aggregate reported zero abstentions while row-level data contained three; (ii) a successor implementation with derived aggregates restores invariant consistency; (iii) a deterministic identity contract passes under synthetic fixtures but a single observed-source pair admits downstream passage when a guard is bypassed; and (iv) historical sequence and copy-number corpora could not be recovered from admitted provenance. A morphology null benchmark is retained as disclosed negative cached evidence. We do not claim therapeutic efficacy, clinical utility, population-level biomedical error rates, or real-world evidence.

1 Introduction

Machine-learning systems for biomedical hypothesis generation must separate *what was computed* from *what may be claimed*. Content-addressed storage and hash verification establish integrity of bytes, not meaning: aggregates can disagree with row ledgers, evaluation wiring can fail to exercise intended disagreements, and missing corpora can leave historical numbers unverified. We frame this work as an evidence-pipeline control problem: normalize, validate, admit or abstain, derive results, and attach claim ceilings before projection to manuscripts [**? ?**].

2 Related Work and Terminology

Selective prediction and reject options allow models to abstain under uncertainty [**? ?**]. Workflow and data provenance standards document lineage without guaranteeing semantic correctness [**?**]. Biomedical identifier normalization relies on authorities such as HGNC and UniProt [**? ?**]. Reproducibility guidance distinguishes replication from reproducibility and encourages reporting null results [**?**]. Regulatory Real-World Evidence frameworks define boundaries we do not claim here [**?**].

We distinguish *cryptographic custody* (hash-valid, content-addressed storage) from *semantic verification* (terminal-state invariants, identity resolution, aggregate-row consistency).

Synthetic fixture. An intentionally constructed input used to test deterministic software behavior.

Canonical contract fixture. A project-defined synthetic or curated fixture whose expected output is fixed before execution and used to test a deterministic contract.

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Source $\rightarrow$ **Normalize / Validate** $\rightarrow$ **Admit** | **Abstain** $\rightarrow$ **Derived artifact** $\rightarrow$ **Claim (ceiling)** $\rightarrow$ **FCG custody**

Cryptographic layer: SHA-256 / content address *Semantic layer:* row invariants, identity contract

Figure 1: Verify-or-abstain evidence pipeline. Custody without semantic checks can admit inconsistent aggregates.

Experiment	Unit	Guard	Condition	N	Ceiling
GAP accounting repair	row	aggregate invariant	successor run	4	semantic audit
G3 contract	fixture	identity	Lane A	synth.	deterministic
G3 source ablation	pair	identity bypass	Lane B	1	descriptive
G1 provenance	corpus	—	not recovered	—	audit only
G2 provenance	corpus	—	not recovered	—	audit only
Morphology null	ranking	shuffle null	cached	50	repurposing hyp.

Table 1: Primary results from frozen experiment artifacts. Historical G1/G2 counts are excluded.

Observed source record. An actual record retrieved from an identified external source rather than invented for testing. Evaluations using such records are *observed-source evaluations*; they are not Real-World Evidence (RWE) unless regulatory RWD/RWE criteria are satisfied [**?**].

External biological evidence. A source-backed biological record or publication used as evidence for a biological assertion.

Abstention / reject option. Following Chow [**?**] and selective-prediction literature [**?**], a pipeline may emit a terminal ABSTAIN state rather than admit an under-supported record.

Closed-vocabulary identity resolution. Map source strings to a frozen alias table with exact, mapped, or unresolved outcomes; unresolved inputs must not silently pass as canonical.

Provenance. Origin and processing lineage of entities [**?**]. Hash equality does not imply semantic consistency between derived artifacts.

3 Verify-or-Abstain Architecture

Figure 1 sketches the pipeline: SOURCE $\rightarrow$ normalize/validate $\rightarrow$ admit or abstain $\rightarrow$ derived result $\rightarrow$ claim with ceiling $\rightarrow$ frozen custody graph. Semantic verification (terminal states, invariants, identity guards) complements cryptographic custody.

4 Experimental Design

Experiments are preregistered where noted; August 13 historical runs are preserved and not retroactively relabeled. EXP-GAP-ACCOUNTING-001.1 repairs aggregate derivation from rows. EXP-004 ablates a closed-vocabulary identity guard (Lane A: contract; Lane B: N=1 observed-source pair). EXP-002/003 provenance audits and external recovery (EXP-002-PROV.1, EXP-003-PROV.1) search admitted custody without rerunning frozen closeouts. EXP-005 preregistration remains locked but corpus/metric incomplete (not executed). EXP-006 documents a cached morphology null benchmark.

5 Results

Table 1 summarizes terminal classifications and claim ceilings from machine-readable receipts.

GAP accounting. Historical summary reported zero abstentions; row-level candidates contained three ABSTAIN terminals. Successor run: N_{input}=4, $N_{admitted}$=1, $N_{abstained}$=3, summary abstain count = 3; invariants pass.

Identity guard. Lane A: deterministic closed-vocabulary contract passes. Lane B: one observed source/canonical pair (`TAZ/cardiolipin module` $\rightarrow$ `TAZ`) yields ABSTAIN under guard and downstream admission under bypass (*unsupported identity admission*); not a population error rate. Historical evaluation invoked `reconcile(symbol, symbol)`; regression confirms the wiring defect.

Provenance recovery. Admitted external bootstrap package (206 MB, SHA-256 verified) does not contain recoverable G1/G2 derivation chains; numeric substring matches in unrelated CSV fields are rejected as evidence.

Null benchmark. Cached evaluation: known-pair rank 28/50, reciprocal-rank enrichment vs. shuffle < 1; disclosed miss not suppressed.

6 Discussion and Limitations

Scope is intentionally small: $N{=}1$ observed-source ablation; G1/G2 inputs not recovered; same designers performed parts of evaluation; no therapeutic efficacy, biological rescue, clinical utility, RWE, or population prevalence claims; EXP-005 replication not executed pending corpus freeze; infrastructure integrity does not establish semantic correctness.

7 Reproducibility and Provenance

Artifacts, preregistrations, and SHA-256 manifests reside in the project Frozen Custody Graph. Anonymous supplementary materials omit identifying metadata. Historical unverified counts remain quarantined as `UNVERIFIED_HISTORICAL_CLAIM`.

8 Conclusion

Hash-valid custody is necessary but insufficient for trustworthy evidence pipelines. Explicit terminal states and invariants surface failure modes that remain syntactically admissible—a prerequisite before any biomedical performance claim.

References

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